

Session Handling in Node.js: A Tutorial

Session handling is a globally used concept, without which any website or app, regardless of whether it is written in Node.js, PHP or any other backend language, will be in a precarious condition. It enables the user information to be persistent in a 'stateful' manner across all the pages of that website or app. In this article, we shall use Node.js to create and manage sessions.



Let us first look at how a standard HTTP request works (without sessions) and what are its drawbacks. Then we shall discuss what a session is, and how it overcomes all the drawbacks and adds to the security of a client-server connection.

The Hypertext Transfer Protocol

The classic Hypertext Transfer Protocol (HTTP) is a stateless tool. This means every request that is sent from a single client is interpreted by the Web server independently and is not related to any other request. There is no inbuilt mechanism for the server to remember a specific user from different multiple requests, which also makes it impossible for the server to know if each request originated from the same user.

In this model, if the server is required to display some user-specific information, the client has to be authenticated with each and every request. Imagine messaging on Facebook and typing in your user name, email and password every single time you send a new message!

The session solution

To overcome this problem, we need to introduce what is called a state-management protocol for websites and apps. A stateful connection is one that enables the server to realise which requests originate from the same user. This is handled by implementing what

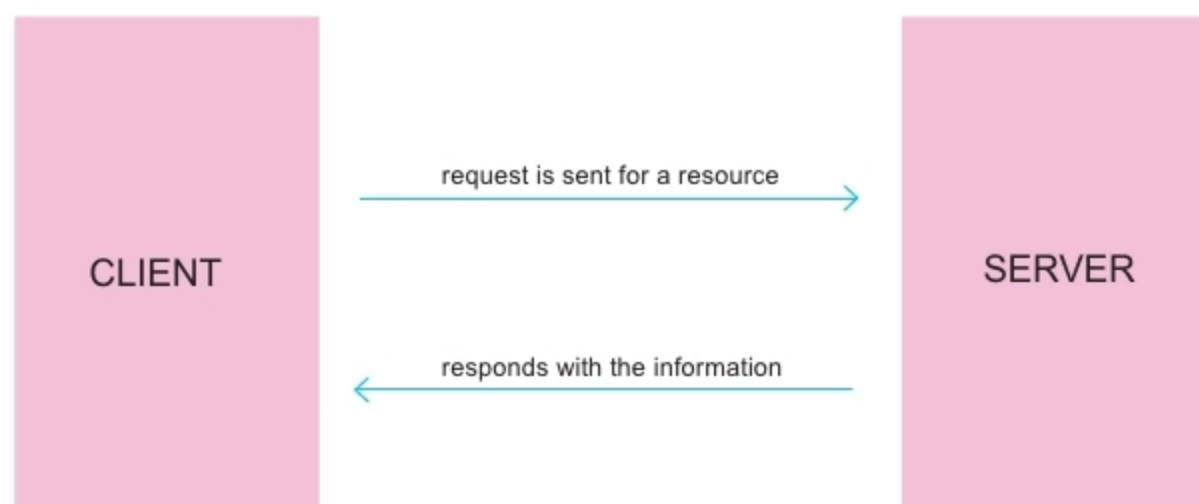


Figure 1: The classic HTTP protocol